

Vibrations Workshop 8 - Periodic Forcing

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ME301

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Resubmission Comment (5/17/26): I fixed the transmissibility plot and the discussion surrounding parameter variations in problem 1. I also added a list of assumptions.

Overview

In this assignment, various systems will be forced via external sinusoidal forcing functions. From there, they will be analyzed in the Laplace domain in which circuit analysis can be applied to *simplify* the dynamics.

Assumptions

Assumptions include, but are not limited to: Newton's Second Law, and the Electrical-Mechanical Analogy pertaining to the loop voltage law, and the use of current generators to model constrained velocities, force generators to model constrained forces, and so on so forth.

Problem 1: Periodic Base Excitation

Consider the driven platform (base) with position coordinate (function) given by:

$$x_{base}(t) = A_{base} \sin\left(\frac{2\pi c}{\lambda}t\right)$$

Attached to this platform is an independent mass-spring-damper system (m, k, R) . The position of the mass relative to the platform is denoted by x . In this regard, the net force on the mass (relative to the ground) is given as:

$$F_{net} = -R(\dot{x} - \dot{x}_{base}) - k(x - x_{base})$$

By Newton's Second Law, $F_{net} = m\ddot{x}$, and so the governing equation of motion can be written as such:

$$\begin{aligned} m\ddot{x} + R\dot{x} + kx &= R\dot{x}_{base} + kx_{base} \\ &= R \frac{d}{dt} \left[A_{base} \sin\left(\frac{2\pi c}{\lambda}t\right) \right] + kA_{base} \sin\left(\frac{2\pi c}{\lambda}t\right) \\ &= \frac{2\pi c}{\lambda} R A_{base} \cos\left(\frac{2\pi c}{\lambda}t\right) + kA_{base} \sin\left(\frac{2\pi c}{\lambda}t\right) \\ &= A_{base} \sqrt{\left(\frac{2\pi c}{\lambda}R\right)^2 + k^2} \sin\left(\frac{2\pi c}{\lambda}t + \tan^{-1}\left(\frac{2\pi cR}{k\lambda}\right)\right) \end{aligned}$$

Upon defining

- $\omega_0 = \frac{2\pi c}{\lambda}$
- $A = A_{base} \sqrt{(R\omega_0)^2 + k^2}$
- $\phi = \tan^{-1}\left(\frac{R\omega_0}{k}\right)$

the EOM can be identified with the forced system described by:

$$m\ddot{x} + R\dot{x} + kx = f(t) = A \sin(\omega_0 t + \phi)$$

It will be assumed moving forward that:

- $m = 130$ kg;
- $R = 47$ kg/s;
- $k = 45000$ N/m;
- $A_{base} = 0.04$ m;
- $\lambda = 4$ m;
- $c = 50/3$ m/s;

Overall, this system can be represented by the following circuit in accordance with Kirchhoff's Voltage Law:

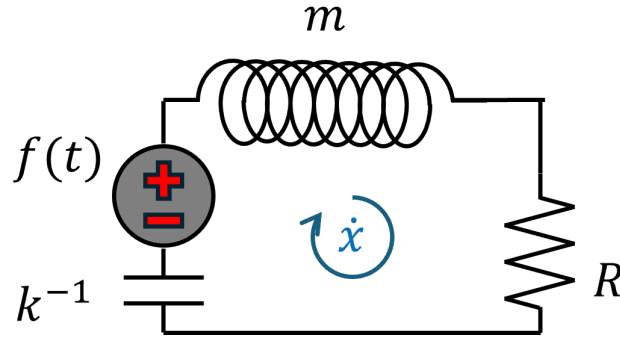


Figure 1: Equivalent Circuit #1

However, it alternatively can be represented as such, where $\dot{x}_b$ is $\dot{x}_{base}$, and V_{base} is its laplace transform:

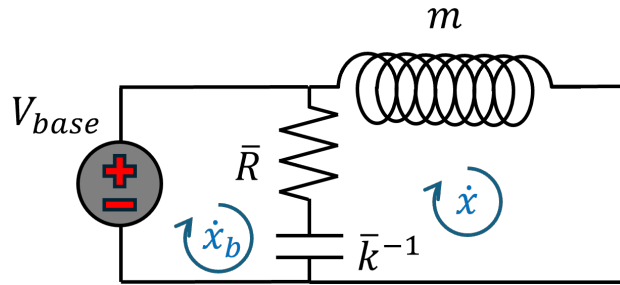


Figure 2: Equivalent Circuit #1 (updated)

$$Z(s) = [ms + R + k/s, -R - k/s; 0, 1] \quad \text{and} \quad F(s) = [0; V_{base}(s)]$$

Thus, the admittance of the system can be determined as given by:

$$Y(s) = Z(s)^{-1}$$

The respective admittance plot (first component) is shown below:

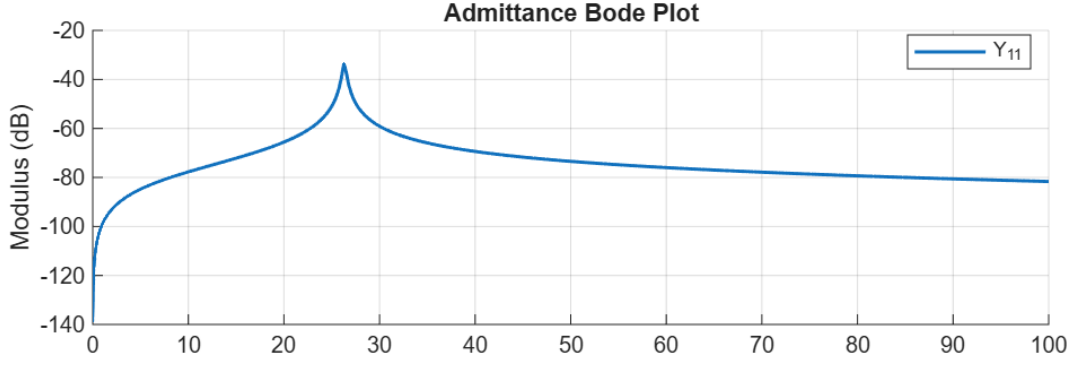


Figure 3: Admittance Magnitude Response #1

Likewise, the FRF response function can be written as:

$$H(s) = Y(s)/s$$

This FRF response can be plotted as such:

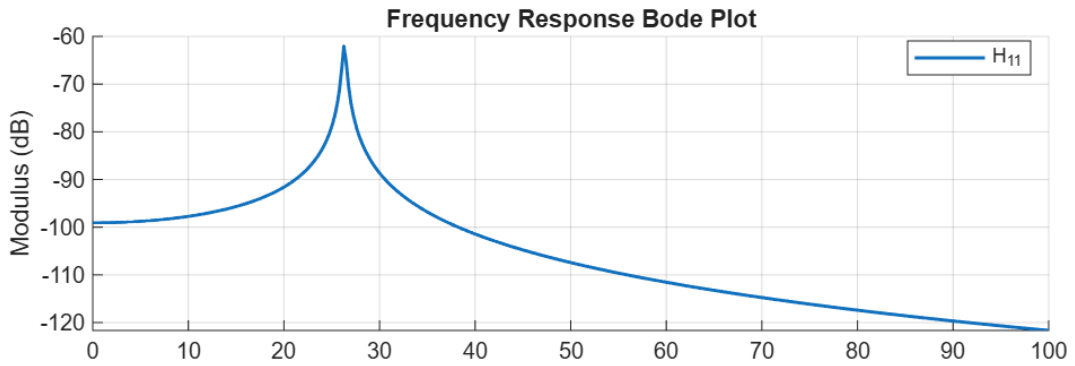


Figure 4: FRF Magnitude Response #1

Because the FRF response is equivalent (in magnitude) to the admittance response divided by the corresponding frequency, the same scaling and shifting properties apply above as they did in the previous plot; the only difference being that each plotted value is scaled down non-uniformly, relatively speaking.

Upon taking the Laplace transform of x_{base} , which is given by

$$X_{base}(s) = \frac{A_{base}\omega_0}{s^2 + \omega_0^2}$$

the transmissibility can be defined in accordance with:

$$T(s) = \frac{X(s)}{X_{base}(s)}$$

The corresponding magnitude plot is shown below:

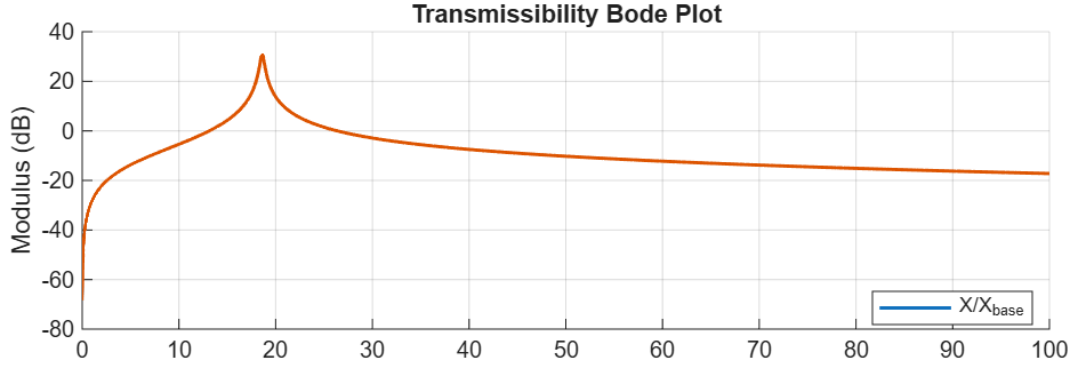
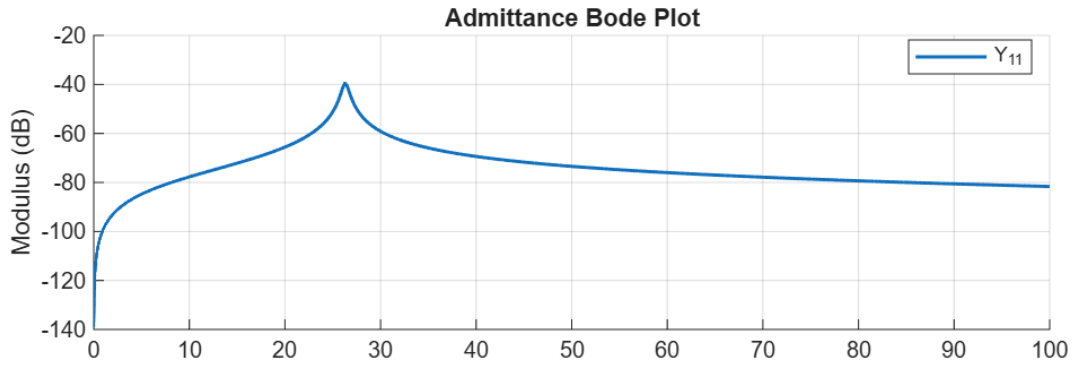
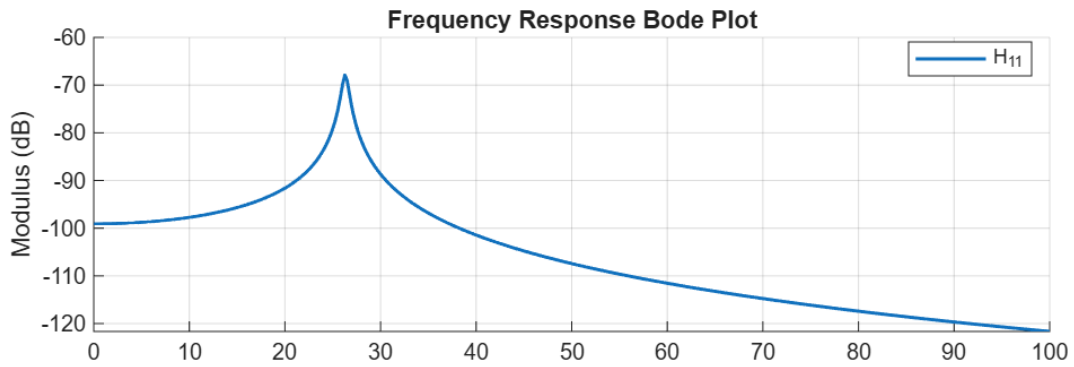


Figure 5: Transmissibility Magnitude Response

Let's now see how changing R , k , and c changes things... To start out, let's increase R .

Figure 6: Admittance Magnitude Response #1 - Increase R

Increasing R visibly decreases the admittance peak without a frequency shift. This just means the system responds less so without compromising the resonate frequency.

Figure 7: FRF Magnitude Response #1 - Increase R

Likewise, increasing R visibly decreases the admittance peak without a frequency shift. The system responds less so without compromising the resonate frequency.

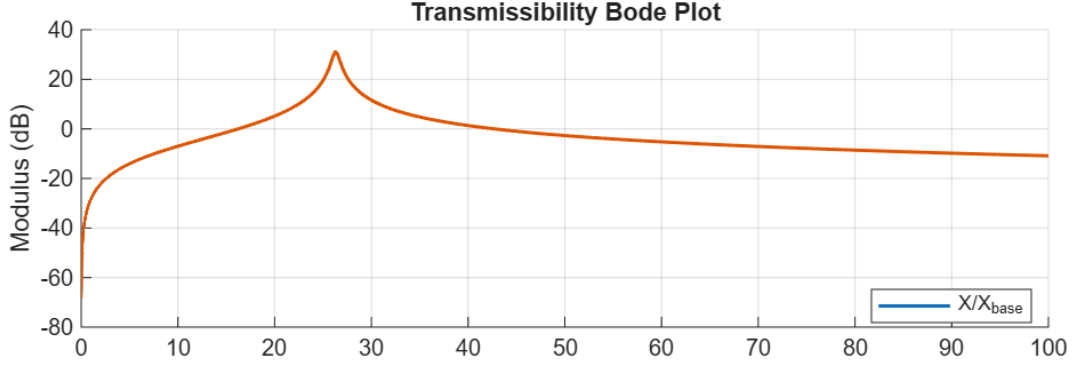


Figure 8: Transmissibility Magnitude Response #1 - Increase R

Here, increasing R lowers the peak value, but shifts the peak frequency rightward. This implies that the system responds less to the motion of the base excitation overall, but even then, the maximum impact the excitation can produce occurs at a higher driving frequency.

Now, let's play around with increasing k .

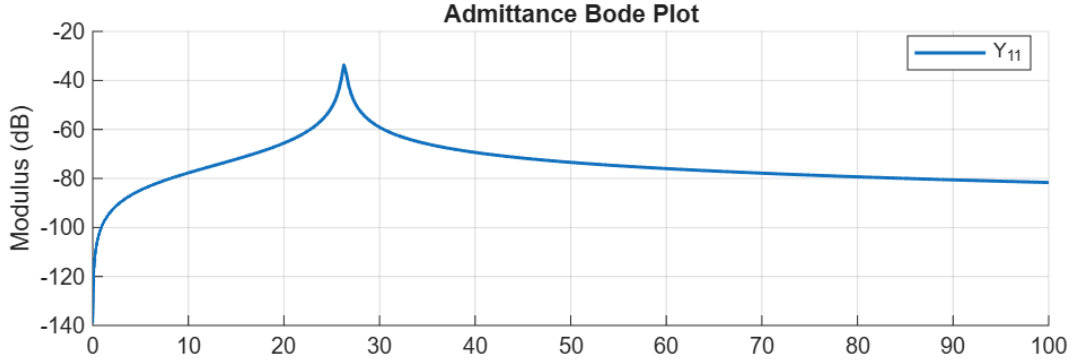


Figure 9: Admittance Magnitude Response #1 - Increase k

Increasing k for the admittance increases both the admittance peak in conjunction with the peak frequency. Therefore, making the system stiffer makes for a more responsive system, but at the cost of requires a higher driving frequency.

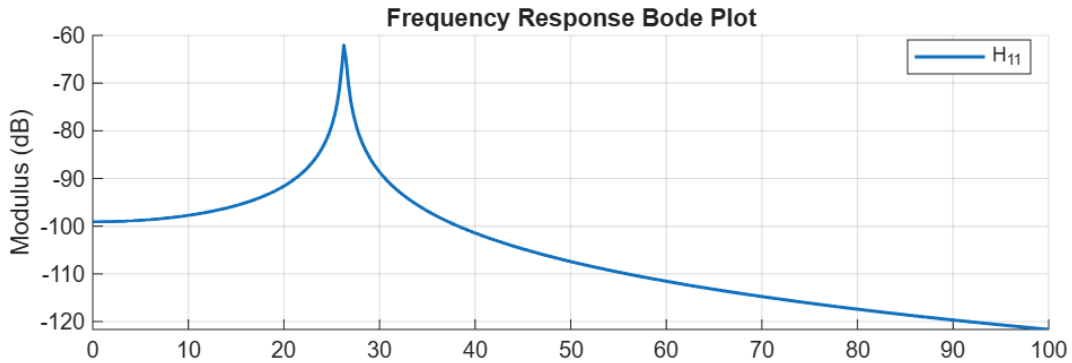


Figure 10: FRF Magnitude Response #1 - Increase k

Likewise, for the FRF, increasing k increases both the peak in conjunction with the peak frequency. So, making the system stiffer makes for a more responsive system, but at the cost of requires a higher driving

frequency.

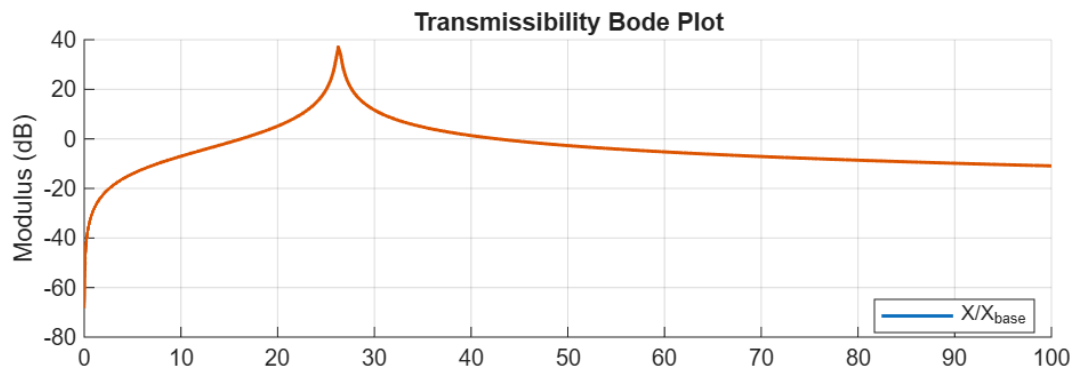


Figure 11: Transmissibility Magnitude Response #1 - Increase k

Interestingly enough, for the transmissibility, increasing k both increases its peak frequency and its peak value. That means increasing the stiffness of the system make the mass respond more to the base excitation, but at the expense of requiring that higher driving frequency.

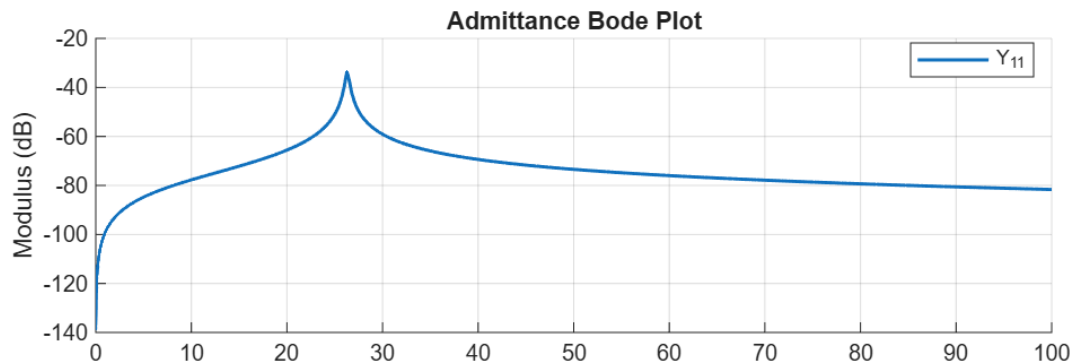


Figure 12: Admittance Magnitude Response #1 - Increase k

Increasing k for the admittance increases both the admittance peak in conjunction with the peak frequency. Therefore, making the system stiffer makes for a more responsive system, but at the cost of requires a higher driving frequency.

Now, let's increase c a bit...

Without even plotting Admittance, the FRF, or the transmissibility, we know the plots won't change because the Admittance does not depend on c .

Finally, the position and velocity responses of the mass will be shown below:

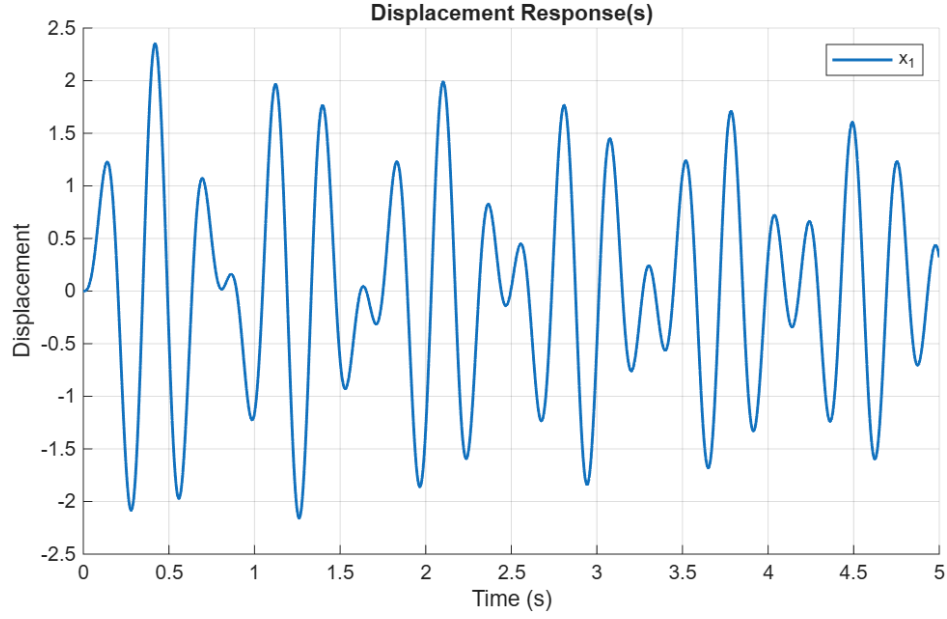


Figure 13: Displacement Response #1

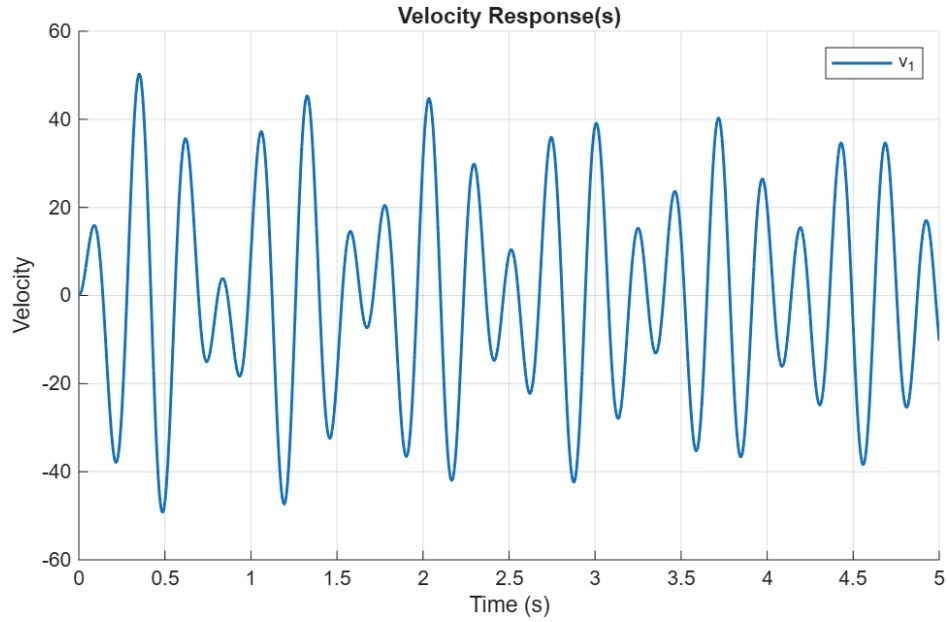


Figure 14: Velocity Response #1

Based off the previous analysis, we expect increasing R to make the system die out faster just as usual until the driving excitation mode dominates. Increasing k will likely do very little to influence the dynamics due to the mild FRF alteration. Lastly, increasing c will almost definitely make the mass oscillate with a greater and greater amplitude up until a point upon which the motion of the mass will match the motion of the platform. This is because the FRF has no dependence on c and so the mass will just go with the flow of the platform.

Problem 2: The Infamous Washing Machine

Consider the following system which encodes the lumped dynamics of clothes spinning around in a washing machine. Note that the rotational inertia of the washing machine (of mass M) will be neglected, and the clothing will be considered to be a point mass m .

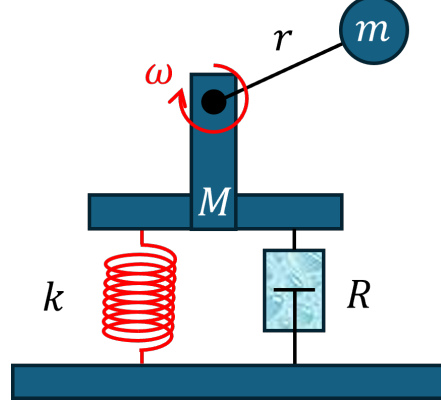


Figure 15: The Washing Machine

One can see that the washing machine is modeled as a mass-spring-damper system. Attached to it is the mass m rotating about (the COM) of M at an angular frequency of ω . Moving forward, let $M = 76$ kg, $m = 4$ kg, $r = 0.25$ m, $k = 60000$ N/m, $R = 10$ kg/s, and $\omega = 85\pi/3$ rad/s.

Provided that the vertical position of M (from equilibrium) is given by x , and that gravity is neglected, the Lagrangian (kinetic minus potential) of the system can be given by:

$$\begin{aligned}\mathcal{L} &= \frac{1}{2}M\dot{x}^2 + \frac{1}{2}m\left\|\frac{d}{dt}(r\cos\omega t, x + r\sin\omega t)\right\|^2 - \frac{1}{2}kx^2 \\ &= \frac{1}{2}M\dot{x}^2 + \frac{1}{2}m(\dot{x}^2 + 2r\omega\dot{x}\cos\omega t + r^2\omega^2) - \frac{1}{2}kx^2 \\ &= \frac{1}{2}(M+m)\dot{x}^2 + mr\omega\dot{x}\cos\omega t + \frac{1}{2}mr^2\omega^2 - \frac{1}{2}kx^2\end{aligned}$$

The conjugate momentum and conservative force can now be computed as follows:

$$p = \frac{\partial \mathcal{L}}{\partial \dot{x}} = (M+m)\dot{x} + mr\omega\cos\omega t \implies \dot{p} = (M+m)\ddot{x} - mr\omega^2\sin\omega t \quad \text{and} \quad F = \frac{\partial \mathcal{L}}{\partial x} = -kx$$

The non-conservative external force (due to damping) is given by $f = -R\dot{x}$, and so by the (modified) Euler-Lagrange equation (i.e. Newton's Second Law):

$$\dot{p} = F + f \implies (M+m)\ddot{x} - mr\omega^2\sin\omega t = -kx - R\dot{x}$$

This equation can be placed into into a more standard form as follows:

$$m\ddot{x} + \bar{R}\dot{x} + \bar{k}x = \bar{F}(t) \quad \text{with} \quad \bar{R} = \frac{R}{1 + M/m}, \quad \bar{k} = \frac{k}{1 + M/m}, \quad \bar{F}(t) = \frac{mr\omega^2\sin\omega t}{1 + M/m}$$

This equation can also be represented in the Laplace domain as follows (with $V = \mathcal{L}\{\dot{x}\}$):

$$(ms + \bar{R} + \bar{k}/s)V(s) = \frac{mr\omega^3}{(1 + M/m)(s^2 + \omega^2)} \quad \longleftrightarrow \quad Z(s)V(s) = \mathcal{L}\{\bar{F}\}(s)$$

Using this equation as a reference, the corresponding circuit diagram in accordance with Kirchhoff's Voltage Law can be illustrated as follows:

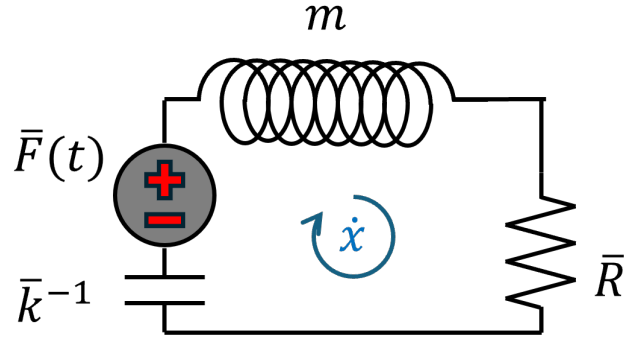


Figure 16: Equivalent Circuit #2

The FRF will now be plotted as instructed:

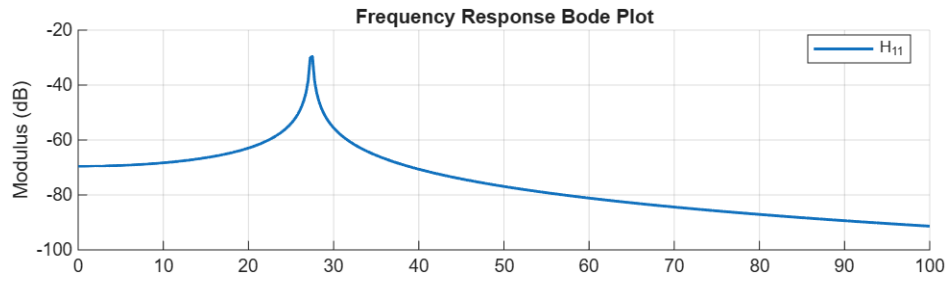


Figure 17: FRF Magnitude Response #2

Nothing out of the ordinary here... Moving on, the next items on the list to plot are the position & velocity responses, and they are shown just below:

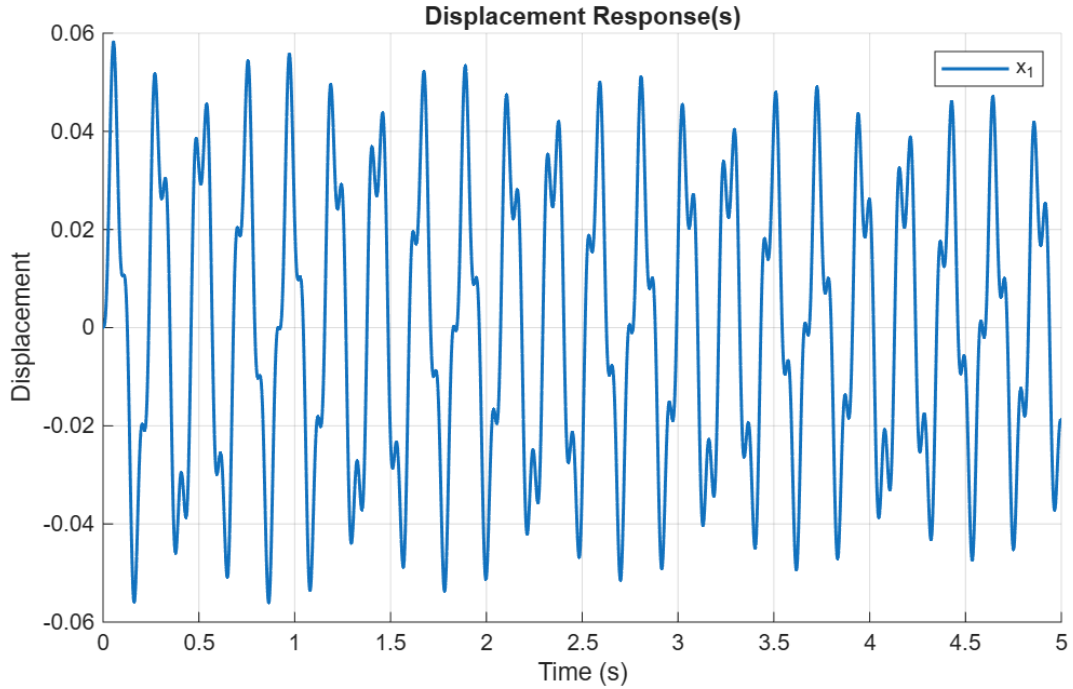


Figure 18: Displacement Response #2

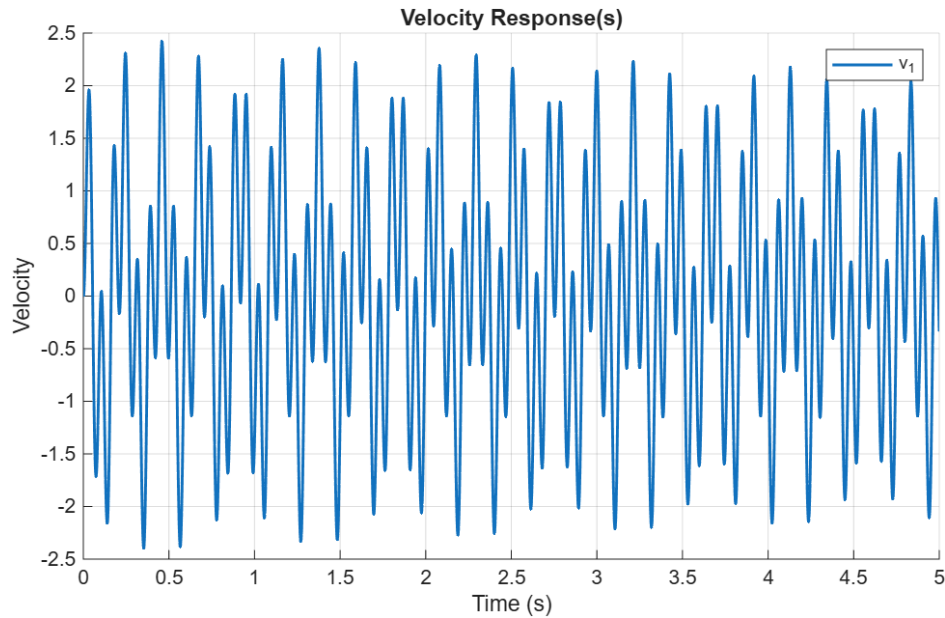


Figure 19: Velocity Response #2

From these plots, it appears as though the washing machine is vibrating rather vigorously! It seems to be jumping up and down around five times per second between 0.04 m and 0.05 m. I certainly would not want that in my pantry, but if it works, then don't fix what's not broken I guess.

Conclusionary Remarks

In this assignment various systems such as base excitation models and washing machines were analyzed. They were modeled respectively as sinusoidally forced mass-spring-dampers. Using techniques from analytical mechanics and circuit analysis, the respective trajectories pertaining to these systems were numerically calculated in MATLAB, and may it be added that the plots are truly immaculate. Not much else can be said except for that this was physics doing its thing on paper.

Also, as the author, I will admit that I'm aware of the fact that velocity generators/current sources were used in lecture to model these kinds of systems. Although I did not use them (because I have the tendency of being lazy), I do know how they work, and I do know how to implement them. So don't worry about that; I happen to like using force generators over them in these simple examples anyways.

Appendix: MATLAB Code

```

1  clc; clear; close all;
2
3  %% Parameters
4  m = 130;
5  R = 47;
6  k = 45000*2;
7  A_base = 0.04;
8  lambda = 4;
9  c = 50/3;
10
11 omega = 2*pi*c/lambda;
12 A = sqrt((R*omega)^2+k^2);

```

```

13 phi = atan(R*omega/k);
14
15 % M = 76;
16 % m = 4;
17 % r = 0.25;
18 % k = 60000;
19 % R = 10;
20 % omega = 85*pi/3;
21 %
22 % R = R/(1+M/m);
23 % k = k/(1+M/m)
24
25 %% Impedance Matrix
26 Z = @(s) ...
27     [m*s+R+k/s, -R-k/s;
28      0, 1];
29
30 %% Control System Toolbox Implementation
31 s = tf('s');
32 Z_tf = Z(s);
33
34 %% Laplaced External Force Vector & Initial Conditions
35 F = [0, s*s*A_base/(omega^2+s^2)];
36 %F = [m*r*omega^3/((1+M/m)*(s^2+omega^2))];
37 x0 = [0, 0];
38 v0 = [0, 0];
39
40 %% Time Domain Parameters
41 t0 = 0;
42 t1 = 5;
43 dt = 0.001;
44
45 %% Coefficient Extraction
46 syms z
47 Z_sym = Z(z);
48
49 n = size(Z(1),1);
50 for i = 1
51     for j = 1
52         Zij = Z_sym(i,j);
53         M(i,j) = double(limit(Zij/z,z,+inf));
54         K(i,j) = double(limit(Zij*z,z,0));
55         R(i,j) = double(limit(Zij - M(i,j)*z - K(i,j)/z, z, 1));
56     end
57 end
58
59 % disp('M:'); disp(M);
60 % disp('R:'); disp(R);
61 % disp('K:'); disp(K);
62
63 %% Augmented Force Vector
64 F_aug = F' + M*v0' - (K*x0')/s;
65
66 %% Dependent Laplace Variables

```

```

67 Y = inv(Z_tf);
68 V = Y*F_aug;
69 V = V(1);
70 X = V/s + x0'/s;
71 H = Y/s;
72 Trans = X/( A_base*omega/(omega^2+s^2) );
73
74 %% Time-Domain
75 t = t0:dt:t1;
76 u = zeros(size(t));
77 u(1) = 1/(t(2)-t(1));
78 [v_resp, t_out] = lsim(V, u, t);
79 [x_resp, ~] = lsim(X, u, t);
80
81 % Extract signals
82 v = squeeze(v_resp(:, :, 1));
83 x = squeeze(x_resp(:, :, 1));
84
85 %% Plot Velocities
86 figure; hold on; grid on;
87 for k = 1
88     plot(t_out, v(:,k), 'LineWidth',1.4);
89 end
90 xlabel('Time (s)');
91 ylabel('Velocity');
92 title('Velocity Response(s)');
93 legend(arrayfun(@(k) sprintf('v_%d',k),1:n,'UniformOutput',false));
94
95 %% Plot Displacements
96 figure; hold on; grid on;
97 for k = 1
98     plot(t_out, x(:,k), 'LineWidth',1.4);
99 end
100 xlabel('Time (s)');
101 ylabel('Displacement');
102 title('Displacement Response(s)');
103 legend(arrayfun(@(k) sprintf('x_%d',k),1:n,'UniformOutput',false));
104
105 %% BODE Plot (Y)
106 w = logspace(-2,2,1000);
107
108 figure; hold on; grid on;
109 legend_entries = {};
110
111 for i = 1
112     for j = 1
113
114         [mag,phase] = bode(Y(i,j), w);
115
116         mag = squeeze(mag);
117         phase = squeeze(phase);
118
119         subplot(2,1,1); hold on; grid on;
120         semilogx(w,20*log10(mag), 'LineWidth',1.4);

```

```

121         ylabel('Modulus (dB)');
122         title('Admittance Bode Plot');
123
124         subplot(2,1,2); hold on; grid on;
125         semilogx(w,phase,'LineWidth',1.4);
126         ylabel('Phase (deg)');
127         xlabel('\omega (rad/s)');
128
129         legend_entries{end+1} = sprintf('Y_{%d%d}',i,j);
130     end
131 end
132
133 subplot(2,1,1); legend(legend_entries,'Location','best');
134 subplot(2,1,2); legend(legend_entries,'Location','best');
135
136 %% BODE Plot (H)
137 figure; hold on; grid on;
138 legend_entries = {};
139
140 for i = 1
141     for j = 1
142
143         [mag,phase] = bode(H(i,j),w);
144
145         mag = squeeze(mag);
146         phase = squeeze(phase);
147
148         subplot(2,1,1); hold on; grid on;
149         semilogx(w,20*log10(mag),'LineWidth',1.4);
150         ylabel('Modulus (dB)');
151         title('Frequency Response Bode Plot');
152
153         subplot(2,1,2); hold on; grid on;
154         semilogx(w,phase,'LineWidth',1.4);
155         ylabel('Phase (deg)');
156         xlabel('\omega (rad/s)');
157
158         legend_entries{end+1} = sprintf('H_{%d%d}',i,j);
159     end
160 end
161
162 subplot(2,1,1); legend(legend_entries,'Location','best');
163 subplot(2,1,2); legend(legend_entries,'Location','best');
164
165 %% Transmissibility Function
166 figure; hold on; grid on;
167 legend_entries = {};
168
169 [mag,phase] = bode(Trans,w);
170
171 mag = squeeze(mag);
172 phase = squeeze(phase);
173
174 subplot(2,1,1); hold on; grid on;

```

```

175     semilogx(w,20*log10(mag), 'LineWidth',1.4);
176     ylabel('Modulus (dB)');
177     title('Transmissibility Bode Plot');
178
179     subplot(2,1,2); hold on; grid on;
180     semilogx(w,phase, 'LineWidth',1.4);
181     ylabel('Phase (deg)');
182     xlabel('\omega (rad/s)');
183
184     legend_entries{end+1} = sprintf('X/X_{base}');
185
186     subplot(2,1,1); legend(legend_entries, 'Location', 'best');
187     subplot(2,1,2); legend(legend_entries, 'Location', 'best');

```